

USE CASE SPECIFICATION

Online Internship & Placement Prep Portal

UC-1: Register & Login

Actor(s)	Student, Admin
Goal	Allows a new user to create an account and an existing user to authenticate and reach the dashboard for their role.
Pre-condition	The user has access to the portal's landing page. For login, the user already has a registered account.
Main Flow	1. User navigates to the portal's landing page. 2. User selects Register (new user) or Login (existing user). 3. For registration, user enters name, email, password, and role-specific details (e.g., college/batch for students). 4. System validates input format (email syntax, password strength). 5. System checks that the email is not already registered. 6. System creates the account and stores the password in hashed form. 7. System redirects the user to the login page with a success message. 8. User enters email/username and password on the login page. 9. System validates credentials against the stored hashed values. 10. System creates an authenticated session and redirects the user to their respective dashboard (Student or Admin).
Alternate Flow	Invalid Login Credentials At Step 9, if the credentials do not match: 9a. System displays an Invalid email or password message. 9b. System returns the user to the login form (Step 8) without creating a session. Duplicate Registration Email At Step 5, if the email is already registered: 5a. System displays an Account already exists message. 5b. System offers a link to the login page instead of creating a new account.
Post-condition	User is registered (if new), authenticated, and directed to the appropriate Student or Admin dashboard.

UC-2: View Internship Listings

Actor(s)	Student
Goal	Allows a student to browse, filter, and view details of active internship listings posted by the admin.
Pre-condition	The student is logged in (authenticated).
Main Flow	1. Authenticated student navigates to the Internships section. 2. System retrieves active internship listings posted by admin from the database. 3. System displays listings with title, company, and short description. 4. Student applies filters (location, domain, stipend, duration) as needed. 5. System re-queries and refreshes the listing view based on the filters. 6. Student selects a specific listing to view its details. 7. System displays the full description, eligibility, deadline, and current application status (if already applied). 8. Student may bookmark/save a listing for later reference. 9. Student navigates back to the full listings page. 10. Student proceeds to apply directly from the detail view, triggering the Apply for Internship use case.
Alternate Flow	No Listings Available At Step 3, if no active listings exist: 3a. System displays No internships currently available. Filter Returns No Results At Step 5, if the filter criteria match no listings: 5a. System displays No matching internships found. 5b. System lets the student reset the filters.
Post-condition	Student has viewed the listing details and, optionally, saved a listing or proceeded to apply.

UC-3: Apply for Internship

Actor(s)	Student
Goal	Allows a student to submit an application, with resume and cover note, for an active internship listing. (Includes: View Internship Listings)
Pre-condition	The student is logged in and the internship listing is active.
Main Flow	1. Student views internship listings (include: View Internship Listings). 2. Student selects a listing and clicks Apply. 3. System checks whether the student has already applied to this listing. 4. System displays the application form (resume upload, cover note, etc.) if not already applied. 5. Student fills in / reviews the required application details. 6. Student submits the application. 7. System validates that all mandatory fields and attachments are present. 8. System records the application with a timestamp and an Applied status. 9. System updates the listing's application count for the admin view. 10. System displays a confirmation and adds the entry under My Applications.
Alternate Flow	Duplicate Application Blocked At Step 3, if the student has already applied: 3a. System blocks the action. 3b. System displays You have already applied to this internship instead of showing the form. Incomplete Application At Step 7, if mandatory fields are missing: 7a. System highlights the missing fields. 7b. System returns to Step 5 without submitting.
Post-condition	Application is recorded with "Applied" status, listed under My Applications, and visible to the admin.

UC-4: Practice Questions (Aptitude / MCQ / Coding)

Actor(s)	Student
Goal	Allows a student to practise questions by category and receive immediate feedback, with the session results stored for analytics.
Pre-condition	The student is logged in and the question bank contains questions for the chosen category.
Main Flow	1. Student navigates to the Practice section. 2. System displays available categories (Aptitude, Technical MCQ, Coding). 3. Student selects a category. 4. System retrieves questions from the categorized question bank matching that category. 5. System displays the first question with answer options / code editor as applicable. 6. Student submits an answer for the current question. 7. System evaluates the answer and shows immediate correctness feedback. 8. Student proceeds to the next question (Steps 6–7 repeat until the set is exhausted or the student stops). 9. System compiles the practice session score once the student completes or ends the set. 10. System stores the session results and updates the student's practice history for analytics.
Alternate Flow	No Questions in Category At Step 4, if the question bank has no questions for the selected category: 4a. System displays No questions available in this category yet. 4b. System returns to Step 2. Student Exits Mid-Session At Step 8, if the student exits before completing the set: 8a. System saves partial progress. 8b. System stores results up to the last answered question instead of discarding the session.
Post-condition	Practice session score and results are stored in the student's practice history.

UC-5: Take Timed Mock Test

Actor(s)	Student
Goal	Allows a student to attempt a timed mock test and have it scored and stored in their history. (Includes: Auto-Submit on Timeout, View Score & Answer Analysis)
Pre-condition	The student is logged in and at least one mock test is available.
Main Flow	1. Student navigates to the Mock Tests section. 2. Student selects a mock test (by category, difficulty, or duration). 3. System displays the test instructions and time limit. 4. Student clicks Start Test and the system starts the countdown timer. 5. System presents questions with navigation controls to move between them. 6. Student answers questions, marking/reviewing/changing answers within the time limit. 7. Student clicks Submit Test before time expires. 8. System locks further input and records all submitted answers. 9. System calculates the score and correct/incorrect breakdown (include: View Score & Answer Analysis). 10. System stores the test attempt in the student's history for analytics.
Alternate Flow	Time Expires Before Submission At Step 7, if the timer reaches zero before manual submission: 7a. System automatically submits the test using the recorded answers (include: Auto-Submit on Timeout). 7b. System proceeds to Step 8. Connectivity Loss During Test At Step 6, if the student's connection drops: 6a. System preserves the already-recorded answers. 6b. System resumes the same session with the remaining time when the student reconnects, instead of restarting the test.
Post-condition	Test attempt is submitted, scored, and stored in the student's test history.

UC-6: Auto-Submit on Timeout

Actor(s)	System (triggered on behalf of Student)
Goal	Automatically submits a mock test when the time limit expires so that the attempt is saved and scored fairly.
Pre-condition	A mock test attempt is in progress and the countdown timer is running.
Main Flow	1. System continuously tracks elapsed time against the test's configured duration. 2. Timer reaches zero while the student is still attempting the test. 3. System immediately disables all answer input controls. 4. System captures the current state of all answered and unanswered questions. 5. System marks the test attempt status as Auto-Submitted. 6. System records the submission timestamp as the configured end time. 7. System processes the recorded answers the same way as a manual submission. 8. System notifies the student on-screen that the test was auto-submitted due to timeout. 9. System proceeds to scoring and analysis generation. 10. System logs the auto-submit event for audit and analytics purposes.
Alternate Flow	Student Had Just Manually Submitted At Step 2, if the attempt already has a "Submitted" status when the timer expires (race condition): 2a. System skips the auto-submit process. 2b. System avoids creating a duplicate record.
Post-condition	Test attempt is saved with "Auto-Submitted" status and sent for scoring, and the event is logged.

UC-7: View Score & Answer Analysis

Actor(s)	Student
Goal	Allows a student to see the score, accuracy, and a question-by-question analysis of a completed test attempt.
Pre-condition	The student is logged in and has submitted at least one mock test attempt.
Main Flow	1. Student completes/submits a mock test, or navigates from My Test History. 2. System retrieves the recorded answers for the selected attempt. 3. System compares each recorded answer against the correct answer key. 4. System computes the total score, accuracy percentage, and time taken. 5. System displays a results summary (score, rank/percentile if applicable). 6. Student clicks into Detailed Analysis. 7. System displays a question-by-question breakdown (correct / incorrect / skipped) with explanations. 8. Student filters the breakdown by topic or difficulty. 9. Student reviews individual questions and their correct solutions. 10. Student navigates back to test history or proceeds to the analytics dashboard.
Alternate Flow	Analysis Not Yet Generated At Step 3, if scoring is still being computed (e.g., delayed evaluation for coding questions): 3a. System displays Results are being processed. 3b. Student can check back later.
Post-condition	Student has reviewed the score summary and detailed answer analysis for the attempt.

UC-8: View Personal Analytics Dashboard

Actor(s)	Student
Goal	Allows a student to track performance trends and identify strengths and weaknesses across tests and practice sessions. (Includes: View Score & Answer Analysis)
Pre-condition	The student is logged in.
Main Flow	1. Student navigates to the Analytics / Dashboard section. 2. System retrieves the student's full test and practice history. 3. System aggregates scores across attempts to compute performance trends over time. 4. System identifies strength and weakness areas by topic/category. 5. System renders trend charts (e.g., score over time, accuracy by category). 6. Student drills down into a specific test attempt for full details (include: View Score & Answer Analysis). 7. Student filters the dashboard by date range or test type. 8. System recalculates and refreshes the visualizations based on the filter. 9. Student views recommended focus areas generated from the weakness analysis. 10. Dashboard auto-updates the next time the student completes a new test or practice session.
Alternate Flow	Insufficient History At Step 3, if the student has no completed tests or practice sessions yet: 3a. System displays Complete a test or practice session to see your analytics instead of charts.
Post-condition	Student has viewed up-to-date performance trends and recommended focus areas.

UC-9: Manage Internship Listings

Actor(s)	Admin
Goal	Allows an admin to add, edit, and deactivate internship listings that are shown to students.
Pre-condition	The admin is logged in.
Main Flow	1. Admin navigates to Manage Internships. 2. System displays existing listings with status (active / expired / draft). 3. Admin selects Add New Listing. 4. Admin enters listing details (title, company, description, eligibility, deadline, stipend). 5. Admin submits the listing. 6. System validates required fields and deadline validity (must be a future date). 7. System saves the listing and marks it Active, making it visible to students. 8. Admin selects an existing listing to edit its details. 9. Admin saves changes; system updates the listing and timestamps the modification. 10. Admin deactivates/removes a listing; system hides it from student view while retaining historical application records.
Alternate Flow	Validation Failure At Step 6, if required fields are missing or the deadline is in the past: 6a. System rejects the submission and highlights the errors. 6b. System returns to Step 4. Removing a Listing with Active Applications At Step 10, if the listing has pending student applications: 10a. System warns the admin. 10b. System asks for confirmation before deactivating the listing.
Post-condition	Listings are added, updated, or deactivated and students see only the active ones.

UC-10: Manage Categorized Question Bank

Actor(s)	Admin
Goal	Allows an admin to add, edit, and delete categorized questions used for practice and mock tests.
Pre-condition	The admin is logged in.
Main Flow	1. Admin navigates to the Question Bank section. 2. System displays existing categories (Aptitude, Technical MCQ, Coding) with question counts. 3. Admin selects a category or creates a new one. 4. Admin selects Add Question. 5. Admin enters question content, options/test cases, correct answer, and difficulty level. 6. Admin submits the question. 7. System validates that all required fields (e.g., correct answer, options for MCQ) are present. 8. System saves the question to the selected category, making it available for practice and tests. 9. Admin edits or deletes existing questions from the bank. 10. System updates the question bank and reflects the change immediately.
Alternate Flow	Incomplete Question Data At Step 7, if a required field (e.g., no correct answer marked) is missing: 7a. System rejects the submission. 7b. System returns to Step 5 with the error highlighted. Deleting a Question in Use At Step 9, if the question is part of an active or scheduled mock test: 9a. System warns the admin before allowing the deletion.
Post-condition	Question bank is updated and the changes are available to practice sessions and mock tests.

UC-11: View Student Applications

Actor(s)	Admin
Goal	Allows an admin to review student applications across all listings and update their status.
Pre-condition	The admin is logged in.
Main Flow	1. Admin navigates to the Applications section. 2. System retrieves all applications submitted across all internship listings. 3. System displays applications in a table with student name, listing, and status. 4. Admin filters applications by listing, date, or status. 5. System refreshes the table based on the filter. 6. Admin selects a specific application to view full details. 7. System displays the student's submitted resume/cover note and profile information. 8. Admin updates the application status (e.g., Shortlisted, Rejected, Selected). 9. System saves the status change and timestamps it. 10. System reflects the updated status to the student under My Applications.
Alternate Flow	No Applications Yet At Step 3, if no student has applied to any listing: 3a. System displays No applications received yet. Bulk Status Update At Step 8, if the admin wants to update several applications at once: 8a. Admin selects multiple applications and applies a status change in bulk. 8b. System validates and updates all selected records in a single operation.
Post-condition	Application status is updated, timestamped, and visible to the student.